CANALRUNG · DESIGN DECISIONS REGISTER

CanalRung design decisions register

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.3	2026-11-05	Amish Chadha	Kit 1.9.1: concept choice opened (CNR-DDR-004); two option B fixes recorded as decided under the 2026-10-23 standing instruction
0.2	2026-10-03	Amish Chadha	Amish's requirement decisions 4A and 9A of 2026-10-03 recorded (CNR-DDR-003): 9 mm ropes with R2 restated to 4.5 kg; the 500 x 1,200 mm footing standard at every site
0.1	2026-10-03	Amish Chadha	Register opened at TRL 3; every decision made under Amish's 2026-10-03 pre-approval

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Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

> **Safety:** CanalRung is safety-critical rescue and rope equipment. Every decision below that touches safety took the conservative option, and the decision records say what evidence would relax it. Nothing here makes the ladder safe to use without proof loads and supervised trials (TRL 4). Never enter the water to rescue someone.

Open decisions

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Which concept option to take forward (Proposed, awaiting Amish)	A thrown rope ladder; B coping-hook folding ladder; C post winch with a floating sling	A (107 of 145 against 92 for C and 88 for B)	Everything: the model, drawing, BOM, calculations and build plan are for A	CNR-DDR-004, CNR-OPT-001
2	Keep USD 605.20 against the USD 500 target as a reported overrun, or take savings (Proposed, awaiting Amish)	Keep and report; take the savings listed under Value engineering	Keep and report; `budget_usd` is not raised	BOM	CNR-CAL-001 I1
3	Footing depth on a slope crest (Proposed, awaiting Amish; independent review finding 1)	Keep 500 x 1,200 mm and rely on a site pull test; deepen to 1,500 mm (saturated 11.6 kN on level ground, factor 2.6; about USD 36 more concrete)	Deepen to 1,500 mm; it changes the standard footing Amish decided in CNR-DDR-003 (9A) and the cost, so it waits for him	Post length (2,300 to 2,600 mm), footing, BOM line 2 and line 1, CAL G4 and G5	CNR-STM-001 S4, S5
4	Whether the R8 ageing allowance applies to the R4 rope factor (Proposed, awaiting Amish; review finding 2)	Factor 4 for a new rope only (now 4.08, aged 3.3); factor 4 aged, which needs a 22 kN rope (type A, heavier and dearer)	Keep the 9 mm rope: aged 3.3 stays above Amish's floor of three; record that R4 means a new rope	Rope, BOM line 19, CAL E1, mass	CNR-STM-001 S3
5	Eye nut pulled off its axis (Proposed, awaiting Amish; review finding 6)	Keep the M16 eye nut and obtain a rating at the worst rope angle; replace it with a swivel hoist ring	Ask the supplier for the off-axis rating first; if none, a swivel hoist ring	Anchor, BOM lines 4 and 5, CAL G1	CNR-STM-001 S23

All other decisions were made under Amish's 2026-10-03 pre-approval, and requirement decisions 4A and 9A were decided by Amish on 2026-10-03 (below).

To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The storage box takes a 370 mm rung across its inside depth (about 440 mm), and its back wall is flat where the plate sits	The folded ladder and the U-bolt clamp depend on it; choose a different box if not	CNR-DDR-002, P6
2	The eye nut is forged and marked with a working load limit of at least 700 kg in line with its thread	The anchor factor of 1.53 rests on it	CNR-CAL-001, G1
3	The screw links are marked with a working load limit of at least 1,000 kg and are long enough (about 60 mm inside) to pass round the eye nut's ring with both rope loops in	The top joint needs room for the ring and two loops	CNR-DDR-002, P7
4	The rope is EN 1891 type B, 9 mm, with a declared breaking strength of at least 18 kN and about 55 g/m; polyester if available	The rope strength of 18.4 kN wet with knots (4.08 times 4.5 kN, R4) and the 4.27 kg ladder mass rest on it	CNR-CAL-001, E1 and B1; CNR-DDR-003
5	The foam tube is closed-cell (a cut end takes up no water) and grips 28.6 mm tube	Buoyancy and the sleeve's fit	CNR-CAL-001, C1
6	The nylon bushes are 12 mm bore, 16 mm outside, with a 22 mm flange, and the 9 mm rope passes freely while its overhand knot cannot pull through	The rope and knot seat	CNR-DDR-002, P1
7	The bank soil at the first site is granular (sand or gravel), drained or saturated, and not soft clay or peat	The standard 500 x 1,200 mm footing is checked for granular soil only; clay or peat needs a site pull test	CNR-CAL-001, G4 and G5; CNR-DDR-003
8	The Rapid Rung patent number, read before public release	Confirms the no-elastic-stowing design-around	CNR-PRC-001

Value engineering

Value-engineering target: USD 500 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 605.20 (USD 105.20 over the target). Main cost drivers and savings worth trying:

- The station costs USD 358.80 and the ladder USD 246.40. The largest lines are the concrete footing (USD 143, the 500 x 1,200 mm footing now standard at every site), the 13 aluminium rungs (USD 72.80), the post (USD 52.60), the storage box (USD 45), the ropes (USD 32) and the edge protectors (USD 30).
- Savings worth trying: ready-mix delivered for several stations at once instead of bagged premix; buying rope by the 50 m reel; a post and footing shared with an existing district marker post or fence line; sewn canvas edge protectors instead of bought ones; rungs cut from 6 m tube lengths.
- Mass: the ladder as thrown is 4.27 kg, inside the 4.5 kg of R2 as restated on 2026-10-03. Worth trying at TRL 4 if the throw trials ask for less: stand-off blocks with a lightening hole (about 0.10 kg), a 0.4 kg throw weight (0.10 kg). Lighter rungs were rejected on strength (CNR-DDR-001, D3), and ropes below 9 mm would fall under Amish's three-times floor (CNR-DDR-003).

Decisions made

Decided by Amish under his pre-approval of 2026-10-03, except the two rows dated 2026-10-03 on R2 and R6 (4A and 9A) and the last two rows (2026-11-05, standing instruction of 2026-10-23). He decided 4A and 9A directly on 2026-10-03: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost."

Date	Decision	Decided by	Record
2026-10-03	Design envelope: water to 2.0 m below the coping on 1.5:1 slopes and to 3.0 m on vertical walls; currents to 2.0 m/s; a longer variant later	Amish, pre-approval quoted above	CNR-DDR-001, D1
2026-10-03	The station post is the anchor: forged M16 eye nut on an M16 bolt through the post, 250 mm above ground, ladder linked in advance	Amish, pre-approval quoted above	CNR-DDR-001, D2
2026-10-03	Rungs of 28.6 x 1.65 mm 6061-T6 tube (strength before mass)	Amish, pre-approval quoted above	CNR-DDR-001, D3
2026-10-03	Ropes of 10.5 mm EN 1891 type A kernmantle, polyester preferred, through nylon bushes; overhand knot under and seizing over each rung end	Amish, pre-approval quoted above	CNR-DDR-001, D4
2026-10-03	Stand-off blocks on rungs 3, 6, 9 and 12, 95 mm off the wall	Amish, pre-approval quoted above	CNR-DDR-001, D5
2026-10-03	Soft 0.5 kg throw weight 280 mm below the bottom rung, rungs floating	Amish, pre-approval quoted above	CNR-DDR-001, D6
2026-10-03	No throw line on the first prototype	Amish, pre-approval quoted above	CNR-DDR-001, D7
2026-10-03	Bought 70 L UV-stabilised box clamped to the post, no lock, drain holes, rope notch, rescue knife	Amish, pre-approval quoted above	CNR-DDR-001, D8 and D11
2026-10-03	Load standard: EN 1891 rope; rung and anchor targets of CNR-REQ-001, proof-loaded with CalRig at TRL 4; no certification claimed	Amish, pre-approval quoted above	CNR-DDR-001, D9 and D12
2026-10-03	Keep the stronger rung and report R2 (mass) as not met	Amish, pre-approval quoted above	CNR-DDR-001, D10
2026-10-03	First candidates to approach: Kittitas Reclamation District (co-design) and Water Safety Council of Fresno County (water safety); not contacted, not agreed	Amish, pre-approval quoted above	CNR-DDR-001, D13
2026-10-03	Pitch, problem and the USD 500 value-engineering target unchanged; the USD 43.60 overrun accepted	Amish, pre-approval quoted above	CNR-DDR-001, D14
2026-10-03	Design for construction: rope-through-rung joint with bushes, knots and seizings; bolted stand-off blocks; foam sleeve lengths; end caps; eye nut with crush tube; box on a plate with U-bolts; rope notch and drain holes; screw links; edge protectors; footing sizes	Amish, pre-approval quoted above	CNR-DDR-002
2026-10-03	Saturated or soft banks get a 500 x 1,200 mm footing and a 2,300 mm post (conservative; a site pull test could relax it)	Amish, pre-approval quoted above	CNR-DDR-002, P11
2026-10-03	Appearance model and render scenes as modelled, with no appearance deviations from the model	Amish, pre-approval quoted above	`docs/REVIEW.md`
2026-10-03	R2 (4A): side ropes downsized to 9 mm EN 1891 type B (18.4 kN wet with knots, 4.08 times 4.5 kN, above the three-times floor);		CNR-DDR-003

Date	Decision	Decided by	Record
	stronger rungs kept; the ladder still exceeds 4.0 kg (4.27 kg), so R2 restated to 4.5 kg. Supersedes the 10.5 mm rope of D4 and the "R2 not met" of D10	Amish, 2026-10-03: "1A 2A 3A 4A 5A 6A 7A 8A 9A 10A 11A"	
2026-10-03	R6 (9A): the 500 x 1,200 mm footing with a 2,300 mm post is the standard footing at every site (factor 3.58 drained, 1.59 saturated); R6 restated for drained or saturated soil. Supersedes the two footing sizes of P11	Amish, 2026-10-03: "1A 2A 3A 4A 5A 6A 7A 8A 9A 10A 11A"	CNR-DDR-003
2026-11-05	Option B rung wall raised from 1.6 to 3.2 mm (25.4 mm outside diameter) after a STEM failure (206 MPa, 0.86 of yield)	Standing instruction of 2026-10-23 (fix that keeps the working principle)	CNR-DDR-004, item 2; CNR-STM-001 S13
2026-11-05	Option B stile splice moved onto the middle wall pad with a 200 mm sleeve after a STEM failure (188 MPa bearing)	Standing instruction of 2026-10-23	CNR-DDR-004, item 3; CNR-STM-001 S16